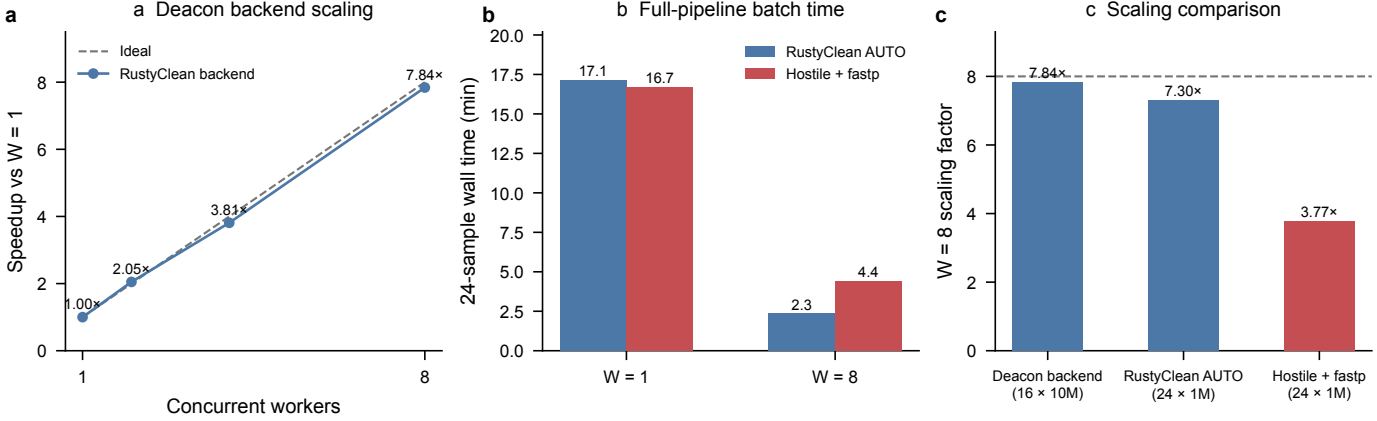
a
a Deacon backend scaling
Ideal
RustyClean backend
Speedup vs W = 1
Concurrent workers
1.00×
2.05×
3.81×
7.84×
b
b Full-pipeline batch time
RustyClean AUTO
Hostile + fastp
24-sample wall time (min)
17.1
16.7
2.3
4.4
W = 1
W = 8
c
c Scaling comparison
W = 8 scaling factor
7.84×
7.30×
3.77×
Deacon backend
(16 × 10M)
RustyClean AUTO
(24 × 1M)
Hostile + fastp
(24 × 1M)